

Math 871 Topology I

Problem Set 5 solutions

1. Show that the product of two Hausdorff spaces is Hausdorff.

Solution. Let $x = (a, b) \in X \times Y$ and $y = (c, d) \in X \times Y$. If $x \neq y$, then $a \neq c$ or $b \neq d$.

Suppose that $a \neq c$. Since X is Hausdorff, we can find neighborhoods U of a and V of c such that $U \cap V = \emptyset$. Thus $U \times Y$ and $V \times Y$ are open in $X \times Y$, and

$$x = (a, b) \in U \times Y \quad \text{and} \quad y = (c, d) \in V \times Y.$$

Moreover, $U \times Y \cap V \times Y = \emptyset$.

Similarly, if $b \neq d$, then we use that Y is Hausdorff and find open neighborhoods U of b and V of d in Y such that $U \cap V = \emptyset$. Then $X \times U$ and $X \times V$ are disjoint and open in $X \times Y$, and

$$x = (a, b) \in X \times U \quad \text{and} \quad y = (c, d) \in X \times V.$$

2. Show that a topological space X is Hausdorff if and only if the **diagonal**

$$\Delta = \{(x, x) \mid x \in X\}$$

is closed in $X \times X$.

Solution. ($\Rightarrow$) Let X be Hausdorff. Given $(a, b) \in X \times X \setminus \Delta$, we have $a \neq b$. Since X is Hausdorff, we can find open sets U and V in X such that $a \in U$, $b \in V$, and $U \cap V = \emptyset$. In particular, $U \times V \cap \Delta = \emptyset$. On the other hand, $(a, b) \in U \times V$, so $U \times V$ is an open neighborhood of (a, b) in $X \times X$ that does not intersect Δ . Since this holds for all $(a, b) \notin \Delta$, we conclude that Δ is closed.

($\Leftarrow$) Suppose that $X \times X$ is closed, and consider distinct points $a, b \in X$. Then $(a, b) \notin \Delta$, and since Δ is closed, there exist a basic open set $U \times V$ where U is open in X and V is open in Y such that

$$(a, b) \in U \times V \quad \text{and} \quad U \times V \cap \Delta = \emptyset.$$

The condition that $U \times V \cap \Delta = \emptyset$ means that for all $(x, y) \in U \times V$, we must have $x \neq y$. In particular, $U \cap V = \emptyset$. Thus U and V are open sets in X such that $a \in U$, $b \in V$, and $U \cap V = \emptyset$. This proves that X is Hausdorff.